

Dataset Overview

We will create a **20,000-patient synthetic clinical dataset** to train machine-learning models that predict Type 2 Diabetes, Prediabetes, Obesity, Metabolic Syndrome, and NAFLD (non-alcoholic fatty liver disease) in a telemedicine platform. The data should mimic a real hospital EHR: medically plausible, multi-label (patients may have multiple conditions), and reflecting clinical diagnostic criteria. Using synthetic data avoids patient privacy issues [31†L117-L125] and allows full control over disease prevalence, feature correlations, and missingness. It also enables reproducibility and easy adjustment of rules (for example, adding new risk factors or changing prevalence) without requiring new IRB approvals [31†L117-L125].

Key goals are: (1) **Medical realism**: follow evidence-based clinical guidelines for all lab values, vital signs, and disease definitions [1†L449-L454] [9†L313-L323]. (2) **Balanced case mix**: reflect a hospital-like population (with significant rates of the target conditions) rather than a healthier community sample. (3) **Machine-learning readiness**: ensure features have known ranges, controlled noise, and manage missing data thoughtfully. We will document all assumptions so that the data can be **automatically generated and validated**.

Note: Because all data are fully synthetic, there are no PHI concerns. We still apply standard ethical considerations: avoid built-in biases (for example, not simulating race unless clinically justified), use medically defensible rules, and document every design choice for transparency.

Disease Background

We target five metabolic-related conditions often managed in primary/specialty care. Each is described below with its clinical features and risk factors.

- **Type 2 Diabetes (T2D)**: A chronic disease of insulin resistance and pancreatic β -cell dysfunction. Adults (often >45 years) develop hyperglycemia marked by high blood glucose and HbA1c [1†L449-L454] [18†L163-L170]. Key risk factors include obesity/overweight, family history of diabetes, sedentary lifestyle, and older age [18†L163-L170]. Major biomarkers are fasting plasma glucose (FPG) and hemoglobin A1c (HbA1c) [1†L449-L454] [18†L163-L170]. T2D is important because it leads to cardiovascular and renal complications; we include it to model glycemic dysregulation in the patient population.
- **Prediabetes**: A state of elevated blood sugar that is not yet in the diabetes range [1†L449-L454]. It is an early warning of insulin resistance. Diagnostic criteria are FPG 100–125 mg/dL or HbA1c 5.7–6.4% [1†L449-L454] [18†L163-L170]. Risk factors largely mirror T2D: overweight/obesity, family history, inactivity, and NAFLD [18†L163-L170]. We include prediabetes because it often precedes T2D and is common (tens of millions in the US [29†L438-L447]), and it can co-occur with other metabolic conditions.

- **Obesity:** A chronic condition of excess body fat, defined by BMI ≥ 30 kg/m² [10†L279-L282] . Causes include diet/environment interactions, genetics, and certain medications [10†L251-L259] . Obesity is a major risk factor for T2D, NAFLD, and Metabolic Syndrome. We follow WHO/CDC definitions: overweight is BMI 25–29.9, obesity ≥ 30 [3†L99-L107] [10†L279-L282] . It is clinically important due to associated hypertension, dyslipidemia, and organ stress. We include obesity as a disease label to capture patients whose BMI/waist values exceed these thresholds [3†L99-L107] [9†L311-L319] .
- **Metabolic Syndrome:** A cluster of metabolic risk factors that together greatly increase cardiovascular risk. It is **diagnosed when ≥ 3** of the following are present: large waist circumference (visceral obesity), high triglycerides, low HDL, elevated blood pressure, and elevated fasting glucose [5†L407-L416] [9†L313-L323] . Underlying causes include insulin resistance, obesity, inactivity, age, and genetics [5†L424-L429] . We include it because it captures the interplay of multiple risk factors. It often coexists with T2D and NAFLD, and signifies high cumulative risk.
- **NAFLD (Metabolic dysfunction-associated steatotic liver disease):** Excess fat accumulation in the liver (steatosis) **not due to alcohol**. It is defined by $>5\%$ – 10% hepatic fat [15†L425-L430] . Risk factors include obesity, insulin resistance, dyslipidemia, and diabetes [15†L455-L458] . Nearly all NAFLD patients have at least one feature of Metabolic Syndrome. We include NAFLD as a label because it is highly prevalent ($\sim 25\%$ of adults [15†L445-L448]) and adds a liver-health dimension to our predictions. There is no simple blood test for NAFLD, so we will assign it based on correlated features (see Section 7).

Each disease’s natural history influences feature values. For example, **early-stage T2D** may only mildly elevate fasting glucose with near-normal A1c, while **severe T2D** often has FPG $>> 180$ mg/dL and HbA1c $> 10\%$. We will simulate mild, moderate, and severe cases by distributing values across clinically observed ranges for each condition.

Clinical Features

We will include the following variables. Each feature is chosen because it is routinely measured and relevant to the target diseases.

- **Patient_ID:** *String*. A unique identifier (e.g. “P00001”). **Unit:** none. **Importance:** Ensures uniqueness. **Mandatory:** Yes. **Availability:** Always (by design). **Validation:** Unique, non-null.
- **Age:** *Integer (years)*. Range ~ 18 – 85 . **Unit:** years. **Clinical Role:** Risk factor for all conditions (older age $\rightarrow$ higher metabolic risk). **Mandatory:** Yes. **Availability:** Always known. **Validation:** ≥ 18 (adult).
- **Gender:** *Category*. Typically “Male” or “Female”. **Mandatory:** Yes. **Availability:** Always. **Validation:** Must be one of the categories. (Sex-specific thresholds apply below.)
- **Height_cm:** *Float (cm)*. Adult height. **Mandatory:** Yes. **Availability:** Almost always measured. **Clinical Role:** Used to compute BMI; slight correlations with risk. **Range:** ~ 145 – 190 cm (female) or 155 – 200 cm (male). **Validation:** Consistency check (weight vs. height for plausible BMI).

- **Weight_kg:** *Float (kg)*. Body weight. **Mandatory:** Yes. **Availability:** Almost always. **Range:** ~40–150 kg. **Validation:** Check BMI bounds after computation.
- **BMI:** *Float (kg/m²)*. Body mass index. **Computed:** from height and weight ($BMI = kg/m^2$). **Mandatory:** Yes (we compute it). **Unit:** none. **Range:** 16–45 (covering underweight to class III obesity). **Importance:** Direct measure for obesity label; strongly related to glucose, lipids, liver enzymes, and blood pressure. **Validation:** Must match Weight/Height ($\pm$ rounding).
- **Waist_Circumference_cm:** *Float (cm)*. Abdominal girth. **Mandatory:** No (often optional). **Availability:** Often measured in metabolic/risk clinics; missing maybe 2–5%. **Range:** ~60–130 cm (women), 65–140 cm (men) [9†L349-L357] . **Role:** Indicator of central obesity; used in Metabolic Syndrome diagnosis. **Validation:** Consistent with BMI (very high waist with low BMI should be rare).
- **Systolic_BP:** *Integer (mmHg)*. Systolic blood pressure. **Mandatory:** Yes. **Range:** ~90–190. **Role:** Elevated BP is both a risk factor and a Metabolic Syndrome component [9†L324-L332] . **Validation:** ≥ 80 and $\geq (Diastolic_BP + \text{few mmHg})$.
- **Diastolic_BP:** *Integer (mmHg)*. Diastolic blood pressure. **Mandatory:** Yes. **Range:** ~60–120. **Validation:** $< \text{Systolic_BP}$.
- **Fasting_Blood_Glucose:** *Integer (mg/dL)*. Fasting plasma glucose (before eating). **Mandatory:** Yes. **Range:** 70–250. **Role:** Primary screening measure for (pre)diabetes [1†L469-L472] . **Validation:** ≥ 0 ; FPG of ~70–99 normal, 100–125 prediabetes, ≥ 126 diabetes [1†L469-L472] .
- **HbA1c:** *Float (%)*. Glycated hemoglobin (2–3 month average glucose). **Mandatory:** Yes. **Range:** 4.5–12.0%. **Role:** Diabetes screening/monitoring [1†L449-L454] . **Validation:** 4.5–5.6 normal, 5.7–6.4 prediabetes, ≥ 6.5 diabetes [1†L449-L454] .
- **LDL_Chol:** *Integer (mg/dL)*. LDL cholesterol (“bad” chol). **Mandatory:** No (often optional in outpatient settings). **Availability:** Common in lipid panels; missing ~2–5%. **Range:** 40–250. **Interpretation:** Optimal < 100 , near-optimal 100–129 [25†L154-L163] . High LDL is a CVD risk factor often co-occurring with metabolic syndrome. **Validation:** ≥ 0 .
- **HDL_Chol:** *Integer (mg/dL)*. HDL cholesterol (“good” chol). **Mandatory:** No. **Availability:** Same as LDL. **Range:** 20–90. **Interpretation:** Men ≥ 40 , women ≥ 50 are considered normal [25†L160-L163] ; lower levels contribute to MetS. **Validation:** ≥ 0 .
- **Triglycerides:** *Integer (mg/dL)*. Blood triglycerides. **Mandatory:** No. **Range:** 40–500. **Interpretation:** Normal < 150 [25†L163-L166] ; 150–199 borderline, 200–499 high; high TG is a MetS criterion [9†L313-L320] . **Validation:** ≥ 0 .
- **ALT:** *Integer (U/L)*. Alanine transaminase, a liver enzyme. **Mandatory:** No. **Availability:** Often ordered if liver disease is suspected. **Range:** 10–200. **Normal:** ~7–55 [23†L462-L470] . Elevated ALT often indicates NAFLD or other liver injury. **Validation:** ≥ 0 .
- **AST:** *Integer (U/L)*. Aspartate transaminase. **Mandatory:** No. **Range:** 10–150. **Normal:** ~8–48 [23†L462-L470] . Elevated AST can accompany ALT in NAFLD. **Validation:** ≥ 0 .

- **Family_History_Diabetes:** *Binary (0/1).* **Mandatory:** No. **Meaning:** 1 if first-degree relative has T2D. **Availability:** Often self-reported; missing in 5–10%. **Role:** Risk factor for diabetes 【18†L163-L170】 . **Validation:** {0,1}.
- **Family_History_Obesity:** *0/1.* Relative with obesity. **Mandatory:** No. **Role:** Genetic risk indicator; missing ~5–10%.
- **Family_History_Hypertension:** *0/1.* Relative with high BP. **Mandatory:** No.
- **Family_History_NAFLD:** *0/1.* Relative with fatty liver. **Mandatory:** No. Family history of NAFLD is less commonly known; missing could be 8–10%.

Notes on “Mandatory vs. Optional”: We treat Age, Gender, Height, Weight, BMI, BP, FPG, HbA1c, and disease labels as mandatory (no missing) in the synthetic training set to simplify modeling. Other features (waist, lipids, liver enzymes, family history) are “optional” (allow missing) since in practice not every patient has those tests or reports those histories.

Clinical Ranges

We define medically accepted ranges for each numerical feature, based on guidelines and physiology:

- **Age:** *Range:* 18–85 years (adult population). (Synthetic dataset restricted to adults.)
- **Height:** Adult human range; we allow ~140–210 cm. (For a 5th–95th percentile, ~150–200 cm.)
- **Weight:** Allow 40–150 kg. (Below 40 kg is extreme adult underweight; above 150 kg is severe obesity.)
- **BMI:** *Categories:* Underweight <18.5, Normal 18.5–24.9, Overweight 25–29.9, Obesity ≥30 【3†L99-L107】 【10†L279-L282】 . *Use:* Diagnose obesity. *Range for dataset:* 16–45.
- **Waist Circumference:** *Thresholds:* Elevated risk starts at ≥94 cm (men) and ≥80 cm (women); obesity at ≥102 cm men, ≥88 cm women 【9†L349-L357】 . *Ranges:* 60–140 cm (female 60–130; male 65–140).
- **Systolic BP:** *Normal:* <120 mmHg 【35†L229-L237】 . *Elevated:* 120–129. *Stage 1 Hypertension:* 130–139. *Stage 2:* ≥140 【35†L229-L237】 . *Range:* 90–190.
- **Diastolic BP:** *Normal:* <80 mmHg 【35†L229-L237】 . *Stage 1:* 80–89. *Stage 2:* ≥90 【35†L229-L237】 . *Range:* 60–120.
- **Fasting Blood Glucose (FPG):** *Normal:* <100 mg/dL 【1†L469-L472】 . *Prediabetes:* 100–125 【1†L469-L472】 . *Diabetes:* ≥126 【1†L469-L472】 . *Range:* 70–250.
- **HbA1c:** *Normal:* <5.7%. *Prediabetes:* 5.7–6.4%. *Diabetes:* ≥6.5% 【1†L449-L454】 . *Range:* 4.5–12%.
- **LDL Cholesterol:** *Optimal:* ~<100 mg/dL; *Near-optimal:* 100–129 【25†L154-L163】 . *High:* ≥160. *Range:* 40–250.
- **HDL Cholesterol:** *Desirable:* ≥40 mg/dL (men), ≥50 mg/dL (women) 【25†L160-L163】 . *Low:* <40 (men)/<50 (women). *Range:* 20–90.
- **Triglycerides:** *Normal:* <150 mg/dL 【25†L163-L166】 . *Borderline:* 150–199. *High:* 200–499. *Range:* 40–500.

- **ALT:** *Normal:* ~7–55 U/L [23†L462-L470] . *Mild elevation:* 56–100. *Significant:* >100. *Range:* 10–200.

- **AST:** *Normal:* ~8–48 U/L [23†L462-L470] . *Range:* 10–150.

These ranges will guide data generation to avoid implausible values (e.g. ALT/AST >200 very unlikely; BMI <16 or >45 extremely rare). We cite official guidelines for the cutoffs: ADA for glucose/HbA1c [1†L449-L454] [1†L469-L472] , CDC/WHO for BMI [3†L99-L107] [10†L279-L282] , AHA/ACC for blood pressure [35†L229-L237] , and CDC for lipids [25†L154-L163] [25†L160-L163] [25†L163-L166] .

Population Design

Age/Gender: We simulate an adult clinic population (18–85). We use the specified age distribution: roughly 20% aged 18–30, 30% in 31–45, 30% in 46–60, 20% in 61–85. Gender is balanced ~50/50. (Older age and equal gender produce a realistic mix.)

BMI Distribution: Obesity is common. We target ~24% obese ($BMI \geq 30$) and a higher percentage overweight (25–29.9), reflecting global trends (WHO: ~43% overweight, 16% obesity worldwide [10†L314-L322]). This yields a mean BMI around 27–30, with some values above 40 (class III obesity).

Blood Pressure: Given age and obesity, we expect many pre-hypertensive/hypertensive patients. We might simulate ~30–40% with $SBP \geq 130$ or on meds, matching guidelines that ~45% of U.S. adults have “high BP” under the new definition [35†L229-L237] .

Glucose/Lipids: Corresponding to the BMI distribution, a substantial fraction will have elevated glucose or dyslipidemia. For example, ~15% may have undiagnosed diabetes-level fasting glucose and ~20–30% prediabetes. We will inject ~10–20% with $LDL > 130$ mg/dL and similar with low HDL, consistent with metabolic risk.

Liver Enzymes: Most patients will have normal ALT/AST; we plan ~2–5% with mild elevation (possible NAFLD) and ~1–2% with moderate/severe elevation.

Dataset Size: We use 20,000 patients (as requested). This is large enough for stable model training and capturing tail patterns. (Academic synthetic-data studies often use similar sizes [31†L117-L125] .)

Justification: The chosen distributions mirror known population statistics (e.g., 12% adult diabetes, 35% prediabetes in the US [29†L438-L447] ; 43% overweight globally [10†L314-L322]) but are tuned to the requested prevalence in Section 9. The mix reflects a clinically realistic sampling rather than a healthy volunteer sample.

Feature Relationships

We will enforce medically valid correlations between features:

- **BMI & Waist Circumference: Very strong positive.** By definition, higher BMI (due to weight) implies larger waist. We will make them nearly deterministic (e.g. $\text{waist} = 0.5 \times \text{BMI} \times 2.5$ with noise) so BMI and waist track closely.
- **BMI & Blood Pressure: Moderate positive.** Higher BMI increases risk of hypertension. We will model SBP/DBP increasing somewhat with BMI (e.g. +1–2 mmHg per BMI point above 25).
- **BMI & Glucose/HbA1c: Moderate positive.** Obesity promotes insulin resistance, so fasting glucose and HbA1c tend to rise with BMI. For example, among high-BMI patients, a larger fraction will have prediabetic or diabetic glucose levels [18†L163-L170] .
- **BMI & Triglycerides (TG): Moderate positive.** Obesity/diet can raise TG. We will correlate TG with BMI modestly.
- **BMI & HDL: Moderate negative.** Higher BMI often associates with lower HDL. We will ensure HDL tends to be lower in high-BMI individuals.
- **Age & Risk Factors: Positive.** Older patients are more likely to have hypertension, diabetes, and Metabolic Syndrome. We will trend SBP, FPG, and disease risk upward with age.
- **Fasting Glucose & HbA1c: Strong positive.** Both measure glycemia. We will ensure consistent labeling: if FPG is in the diabetic range, HbA1c should also generally be high.
- **Triglycerides & HDL: Inverse moderate.** Dyslipidemia often presents with high TG and low HDL simultaneously.
- **Waist & Metabolic Risk:** Larger waist correlates with all MetS factors. We'll often only label MetS=1 if waist is high or several other factors are high.
- **Diabetes & NAFLD: Positive correlation.** NAFLD frequently co-occurs with diabetes (as they share pathophysiology [15†L455-L458] [18†L163-L170]). We will increase NAFLD probability in patients with obesity, high TG, and/or diabetes.
- **Other:** Systolic and diastolic BP correlate strongly (we ensure $\text{SBP} > \text{DBP}$). Male/female differences: e.g., women tend to have higher HDL on average [25†L160-L163] , so we'll add a slight gender effect on HDL.

These relationships ensure that the synthetic population follows known medical patterns: e.g. obese patients tend to be hypertensive and dyslipidemic, diabetic patients tend to have metabolic syndrome features, etc [18†L163-L170] [15†L455-L458] .

Disease Assignment Strategy

Each disease label will be assigned based on multiple features, following clinical criteria with some randomness to simulate variability. We use **multi-factor rules** rather than single cutoffs:

- **Type 2 Diabetes:** Assign if **Fasting Glucose ≥ 126 mg/dL OR HbA1c $\geq 6.5\%$** [1†L449-L454] [1†L469-L472] . In practice, we might not require both, but a patient meeting either threshold will be labeled diabetic. (If both FPG and A1c are borderline high, we may use probability or secondary criteria.) We ensure patients labeled T2D usually have other risk factors (e.g. $\text{BMI} \geq 30$, high TG, hypertension) to reflect severity.

- **Prediabetes:** Assign if **(100 ≤ Fasting Glucose ≤ 125) OR (5.7% ≤ HbA1c ≤ 6.4%)**, and *NOT already labeled T2D* 【1†L449-L454】 【1†L469-L472】 . Prediabetic patients should have mildly elevated glucose/A1c without reaching diabetic thresholds. We'll also simulate that some with borderline lab values but fewer risk factors get Prediabetes.
- **Obesity:** Assign if **BMI ≥ 30** (or equivalently waist ≥102 cm men / ≥88 cm women) 【3†L111-L119】 【9†L349-L357】 . We generate Class I/II/III obesity labels depending on BMI (e.g. 30–34.9, 35–39.9, ≥40) but the main binary label is 1 for any BMI≥30. (This follows WHO/CDC standards 【3†L111-L119】 【10†L279-L282】 .)
- **Metabolic Syndrome:** Assign MetS=1 if **any 3 or more** of the following criteria are met 【5†L407-L416】 【9†L313-L323】 : (1) Waist circumference above threshold (men ≥102 cm or women ≥88 cm) 【9†L349-L357】 ; (2) Triglycerides ≥150 mg/dL 【9†L313-L317】 ; (3) HDL <40 mg/dL (men) or <50 mg/dL (women) 【9†L317-L323】 ; (4) BP ≥130/85 mmHg 【9†L324-L329】 ; (5) Fasting glucose ≥100 mg/dL 【9†L329-L334】 . These are the ATP III/AHA/NHLBI criteria 【9†L313-L323】 . We will ensure at least 3 factors to label MetS. (For example, a patient with central obesity, high TG, and high BP would qualify.)
- **NAFLD (MASLD):** Since we don't have imaging, we assign NAFLD probabilistically based on risk factors. Patients with **obesity** (high BMI), **hypertriglyceridemia**, **elevated ALT/AST**, or existing **diabetes** get higher NAFLD probability. For example, if BMI>28 and TG>180 mg/dL and ALT>50 U/L, label NAFLD=1 with high probability. We also ensure patients with both obesity and T2D often have NAFLD 【15†L455-L458】 【18†L163-L170】 . Conversely, lean patients with normal labs will rarely get NAFLD. (In effect, NAFLD depends on a weighted sum of factors: obesity, dyslipidemia, transaminases, diabetes, age.)
- **Healthy:** Assign Healthy=1 only if **all five disease labels = 0**. Otherwise Healthy=0. This ensures non-overlap (e.g., if any disease is present, Healthy must be 0).

We implement these rules in code so that each synthetic patient's lab/vital values jointly determine the labels. For realism, we add randomness: e.g. a few cases might have BMI 32 but not labeled obesity (measurement error), or borderline MetS factors that randomly count. This avoids deterministic, step-function data.

Disease Co-occurrence

Real patients often have multiple conditions concurrently. Our synthetic data will reflect known co-occurrences:

- **Diabetes and Obesity:** Very common. Most T2D patients will also have obesity (or at least overweight). We will simulate that ~70–80% of labeled diabetics also have BMI>30. (This mirrors the fact that obesity is the strongest risk factor for T2D 【18†L163-L170】 .)
- **Diabetes and Metabolic Syndrome:** Almost all T2D patients meet MetS criteria (type 2 diabetes itself would add one criterion) 【9†L329-L334】 . We will ensure most diabetics also trigger MetS.

- **Obesity and MetS:** Overlap is high. Many obese individuals have high TG, low HDL, or hypertension. We will make sure a large fraction of $BMI \geq 30$ patients are labeled $MetS=1$.
- **NAFLD with MetS/Diabetes:** NAFLD almost never occurs in isolation; it strongly overlaps with MetS components [15†L455-L458] . We will have NAFLD positive patients usually also meet MetS criteria or have T2D. (For instance, >80% of NAFLD cases will coincide with obesity or elevated fasting glucose.)
- **Prediabetes with MetS:** Common. Many prediabetics will also have dyslipidemia or hypertension. We will allow overlap: a patient can be both $Prediabetes=1$ and $MetS=1$ if they meet both rule sets.

These co-occurrences will not be 100% deterministic but will follow probabilities. We can cite that metabolic risk factors tend to cluster: e.g., NAFLD “tends to develop in people who have increased body fat or have diabetes, high cholesterol, or high triglycerides; having a combination of these is known as metabolic syndrome” [15†L455-L458] . In short, **comorbidities will naturally cluster**.

Disease Prevalence

We aim for these approximate prevalences in the **synthetic dataset** (note: these are higher than general-population rates, reflecting a clinical population):

- **Healthy (no diseases): ~30%.** (Majority have at least one metabolic condition.)
- **Prediabetes:** ~18%. (By comparison, in the U.S. ~30–35% of adults have prediabetes [29†L438-L447] ; we target a somewhat lower figure since we deliberately include many chronic cases.)
- **Type 2 Diabetes:** ~16%. (In the U.S., ~12% of adults have diagnosed diabetes [29†L438-L447] . We use ~16% to represent a clinic population with more risk factors.)
- **Obesity:** ~24%. (We target ~24% with $BMI \geq 30$. Globally ~16% of adults are obese [10†L314-L322] ; prevalence is higher in some countries and in health-care settings.)
- **Metabolic Syndrome:** ~18%. (Literature reports varying prevalence (e.g. 14–39% in different countries [26†L6-L10]). Around 20% is reasonable for a synthetic outpatient mix.)
- **NAFLD:** ~12%. (General estimates say ~25% of people have NAFLD [15†L445-L448] ; we use a lower 12% to avoid dominating the dataset, while still allowing NAFLD cases.)

These targets are achieved by adjusting the rules and random sampling. We cite authoritative stats for context: e.g., ADA reports ~12% diabetes and 35% prediabetes in U.S. adults [29†L438-L447] , WHO/CDC report ~43% overweight and 16% obesity in adults [10†L314-L322] , and liver associations estimate ~25% NAFLD prevalence [15†L445-L448] . Our synthetic prevalences are *approximate* and may slightly fluctuate, but we will validate after generation (Section 14) to ensure they are in the desired range.

Borderline Cases

We will include a significant fraction (~15% of patients) with values **near diagnostic thresholds**. This creates “edge” cases for robustness. Examples of borderline values (and few cases around them) include:

- **BMI:** values like 29.5–30.5 (around the obesity cutoff of 30).
- **HbA1c:** values like 5.8–6.5% (prediabetes/diabetes boundary).
- **Fasting Glucose:** e.g. 124–127 mg/dL (just below/above 126).
- **Blood Pressure:** e.g. SBP 128–132 mmHg and DBP 84–86 mmHg (around the 130/85 cutoff).

- **Triglycerides:** e.g. 145–152 mg/dL (near 150).
- **HDL:** e.g. 39–42 mg/dL (men, cutoff 40).
- **Waist Circumference:** e.g. 100–104 cm (male) or 86–90 cm (female) (around MetS criteria).
- **LDL:** e.g. 124–131 mg/dL (near 130 threshold).

Including these borderline patients helps train models to distinguish mild from moderate cases. It also tests that the label assignment logic handles exact-threshold values correctly. In practice, medical decisions often hinge on such borderline labs, so modeling them improves real-world utility. (In contrast, random assignment might yield fewer such cases.)

Noise Model

To mimic real clinical measurements, we will add small random variability to numeric values:

- **Height:** ± 0.5 cm (measurement error or rounding).
- **Weight:** ± 1 kg (scale variation).
- **BMI:** recomputed from noisy height/weight.
- **Blood Pressure:** ± 3 mmHg (typical cuff error) for both SBP and DBP.
- **HbA1c:** $\pm 0.2\%$.
- **Fasting Glucose:** ± 5 mg/dL.
- **LDL:** ± 5 mg/dL.
- **HDL:** ± 3 mg/dL.
- **Triglycerides:** ± 8 mg/dL.
- **ALT/AST:** ± 3 U/L.

These perturbations are small (on the order of normal laboratory or measurement error) so they **break exact ties** but do not blur disease definitions. We will ensure no value goes outside physiologically possible limits after adding noise. Excessive noise is avoided because clinical measurements are relatively precise; we only want to reflect typical lab/imaging variability.

Outlier Strategy

To capture rare extreme cases, we will include a small number (~2%) of outliers with severe but plausible values. Examples of extreme yet valid cases:

- **Extreme obesity:** BMI ~40–45, weight > 120 kg.
- **Severe dyslipidemia:** Triglycerides 300–500 mg/dL, LDL >200 mg/dL.
- **Poorly controlled diabetes:** HbA1c 10–12%, Fasting Glucose >200 mg/dL.
- **Elevated liver enzymes:** ALT ~150–180 U/L, AST ~100–130 U/L.
- **Combined crises:** e.g. BMI 43 *with* HbA1c 10.5% and TG 400.

These cases should remain physiologically plausible. For instance, we would never generate ALT=1000 or LDL=500 (those exceed human limits). We also avoid contradictory outliers (e.g. BMI 18 with 150 kg weight). Outliers increase the model's robustness to rare presentations, but we keep their proportion low.

Missing Data Strategy

In real clinical data, not every feature is measured for every patient. We classify features by expected missingness:

- **Always Available (0% missing):** Patient_ID, Age, Gender, Height, Weight, BMI, SBP, DBP, Fasting_Glucose, HbA1c, and all **disease labels**. These are mandatory in our generated dataset so the model always has core inputs and labels.
- **Usually Available (<5% missing):** e.g. Waist_Circumference and Family_History fields. We simulate a small fraction (1–5%) of these as missing (NULL) to reflect that some clinics may omit waist measurements or patients may not know family history.
- **Occasionally Missing (5–10%):** Liver enzymes (ALT, AST) may be missing in many routine visits. Family_History_NAFLD may be missing up to ~8–10%, since patients often don't know if relatives had fatty liver. We will implement: Waist ~3% missing; LDL, HDL, TG ~2–3% missing; ALT, AST ~5%; Family histories ~5–8%.

Missingness will **not be purely random**. We will induce realistic patterns: for example, younger healthy patients are *less likely* to have extensive labs (so we'll make missing lipid tests more likely if BMI<25 and age<30). Conversely, a patient with obesity/diabetes is more likely to have a full lipid panel. Family history fields will be missing at random or based on age/family size (e.g. older patients more likely to know family history of hypertension).

We will record missing values explicitly as `NULL` and ensure the model can impute or ignore them. (In practice, one could use median imputation or a learned imputer on training data.) Documenting missingness patterns is important for the ML pipeline to handle them appropriately.

Data Validation Rules

After generating each patient, we apply strict validation to catch errors:

- **Value Ranges:** Every numeric value must lie within the specified realistic bounds (see Section 4). E.g. $70 \leq \text{FPG} \leq 250$, $4.5 \leq \text{HbA1c} \leq 12$, $0 \leq \text{LDL} < 300$, $0 \leq \text{AST} < 200$, etc. Negative or physiologically impossible values are rejected.
- **BMI Consistency:** Compute BMI from height and weight; it must match the stored BMI (within rounding). If not, we correct or regenerate the record.
- **Blood Pressure:** Ensure $\text{SBP} \geq \text{DBP}$ and both are ≥ 0 . Also reject $\text{SBP} < 50$ or $\text{DBP} < 30$, which are incompatible with life.
- **Diabetes Labels:** If $\text{T2D}=1$, verify the patient has qualifying labs ($\text{FPG} \geq 126$ or $\text{HbA1c} \geq 6.5$) [1†L449-L454] [1†L469-L472]. If not, fix the label. Similarly, $\text{Prediabetes}=1$ requires FPG 100–125 or HbA1c 5.7–6.4; if the labs are inconsistent, relabel or adjust labs. (We ensure no patient is marked diabetic without lab evidence.)

- **Metabolic Syndrome Label:** If Metabolic_Syndrome=1, check that the patient has at least 3 of the defining criteria [5†L407-L416] [9†L313-L323] . If fewer, we remove the label. Conversely, if 3+ criteria are met, ensure MetS=1.
- **Obesity Label:** If Obesity=1, ensure BMI $\geq$ 30 or waist meets the cutoffs [3†L111-L119] [9†L349-L357] . If not, correct it.
- **NAFLD Label:** Since this is probabilistic, we still verify that NAFLD cases have elevated liver enzymes or severe metabolic features; if a NAFLD label appears for a perfectly healthy profile, it will be flagged for review.
- **Healthy Label:** Enforce Healthy=1 only when all other disease labels are 0. If any disease is 1, Healthy must be 0.
- **Duplicates:** Patient_ID must be unique. We check for duplicate rows and reject or reseed as needed.
- **Cross-checks:** We prevent impossible combos, e.g. BMI=18 with Weight=150 kg (contradiction), or HDL>LDL (while possible, it's unusual if values are normal). If an inconsistency is found (e.g. Weight and Height give BMI 23 but the record lists BMI 35), we either correct it or regenerate.
- **Missing Label Leakage:** No feature should be directly determined by a label. For example, we must ensure **HbA1c or FPG values do not “see” the disease label** – we generate features first, then labels from them, so there is no leakage of the label into the feature values.

All these rules will be coded into the generator's validation step. Any patient failing a rule is either corrected (if easy) or discarded and re-sampled. This ensures the final CSV has no obvious data errors.

Machine Learning Considerations

We prepare the dataset with standard ML best practices:

- **Encoding:** Gender will be binary-coded (e.g. 0/1). All other features are numeric. Disease labels are binary. No additional categorical variables.
- **Scaling:** Since tree-based models (Random Forest, XGBoost, LightGBM) are planned, scaling is not strictly necessary. If we also use logistic regression or neural nets, we could normalize continuous features (e.g. zero-mean/unit-variance). But in a tree ensemble, raw scales (mmHg, mg/dL, etc.) are acceptable.
- **Missing Values:** For model training, missing numeric fields (waist, lipids, ALT/AST) could be imputed (e.g. with the median or mean of training data) or a missing indicator could be added. We will handle missingness carefully – for example, imputing NAFLD risk factors as normal if unknown. Categorical missingness (none in our design except family history flags) could be filled with 0/“None”. Any imputation should be **fit on the training set only** and then applied to validation/test.
- **Feature Engineering:** We already compute BMI from height/weight. We could also create features like “Waist/BMI”, pulse pressure (SBP–DBP), or interactions, but we won't overcomplicate. The chosen raw features themselves (especially lab values and vitals) should suffice for gradient-boosted trees.
- **Class Imbalance:** Some labels (e.g. NAFLD at 12%) are imbalanced. We can use techniques like class weighting or oversampling positive cases in training for those diseases. Evaluation metrics (below) will account for imbalance.
- **Validation:** Use stratified K-fold cross-validation, ensuring each fold has similar disease prevalence. We also reserve a final test set (e.g. 20%) for unbiased evaluation.
- **Evaluation Metrics:** For each disease label (multi-label setting), we will report sensitivity, specificity, precision, recall, F1-score, and ROC-AUC. Overall accuracy is less informative due to imbalance; macro-averaged F1 is better. Precision-recall curves can be used for rare conditions.

- **Explainability:** We will compute feature importances and SHAP values for each model. For example, we expect BMI, fasting glucose, and age to rank high for diabetes predictions. This interpretable analysis will be part of the deliverable.

Final Dataset Blueprint

Feature	Data Type	Unit	Range (min–max)	Mandatory/Optional	Missing %	Related Diseases	Importance
Patient_ID	String	—	P00001... P20000	Mandatory	0%	—	Unique ID
Age	Integer	years	18–85	Mandatory	0%	All	High
Gender	Categorical	—	Male / Female	Mandatory	0%	All	Medium
Height_cm	Float	cm	140–200 (adult)	Mandatory	0%	Obesity, MetS, NAFLD	High
Weight_kg	Float	kg	40–150	Mandatory	0%	Obesity, MetS, NAFLD	High
BMI	Float	kg/m ²	16–45	Mandatory (calc)	0%	Obesity, MetS, Diabetes, NAFLD	High
Waist_Circumference_cm	Float	cm	60–140	Optional	~3%	Obesity, MetS, NAFLD	High
Systolic_BP	Integer	mmHg	90–190	Mandatory	0%	Hypertension, MetS	Medium
Diastolic_BP	Integer	mmHg	60–120	Mandatory	0%	Hypertension, MetS	Medium
Fasting_Blood_Glucose	Integer	mg/dL	70–250	Mandatory	0%	Prediabetes, Diabetes, MetS	High
HbA1c	Float	%	4.5–12.0	Mandatory	0%	Prediabetes, Diabetes	High
LDL_Cholesterol	Integer	mg/dL	40–250	Optional	~2%	MetS, CVD Risk	Medium

Feature	Data Type	Unit	Range (min–max)	Mandatory/ Optional	Missing %	Related Diseases	Importance
HDL_Cholesterol	Integer	mg/dL	20–90	Optional	~2%	MetS, CVD Risk	Medium
Triglycerides	Integer	mg/dL	40–500	Optional	~2%	MetS	Medium
ALT	Integer	U/L	10–200	Optional	~5%	NAFLD	Medium
AST	Integer	U/L	10–150	Optional	~5%	NAFLD	Medium
Family_History_Diabetes	Binary (0/1)	—	0 / 1 (Yes/No)	Optional	~5%	Diabetes	Medium
Family_History_Obesity	Binary	—	0 / 1	Optional	~5%	Obesity	Medium
Family_History_Hypertension	Binary	—	0 / 1	Optional	~5%	Hypertension	Medium
Family_History_NAFLD	Binary	—	0 / 1	Optional	~8%	NAFLD	Medium
Type2_Diabetes	Binary	—	0 / 1	Mandatory (target)	0%	All metabolic markers	High (label)
Prediabetes	Binary	—	0 / 1	Mandatory (target)	0%	Glucose/A1c mildly high	High (label)
Obesity	Binary	—	0 / 1	Mandatory (target)	0%	BMI, Waist	High (label)

Feature	Data Type	Unit	Range (min-max)	Mandatory/Optional	Missing %	Related Diseases	Importance
Metabolic_Syndrome	Binary	—	0 / 1	Mandatory (target)	0%	WC, BP, TG, HDL, Glucose	High (label)
NAFLD	Binary	—	0 / 1	Mandatory (target)	0%	Obesity, TG, ALT/AST, T2D	Medium (label)
Healthy	Binary	—	0 / 1	Mandatory (target)	0%	—	High (label)

(“Importance” ranks features for prediction; e.g. Age/Fasting_Glucose/BMI are High for diabetes modeling. “Validation” notes key rules: see sections above and cited guidelines.)

References

- ADA Standards and NIH Guidelines for diabetes diagnosis [1†L449-L454] [1†L469-L472] .
- CDC/NIH on diabetes risk factors [18†L163-L170] .
- CDC/WHO obesity definitions [3†L99-L107] [10†L279-L282] .
- AHA/ACC definitions of metabolic syndrome and hypertension [5†L407-L416] [9†L313-L323] [35†L229-L237] .
- CDC guidelines for cholesterol and triglycerides [25†L154-L163] [25†L160-L163] [25†L163-L166] .
- Mayo Clinic normal ALT/AST ranges [23†L462-L470] .
- Liver Foundation on NAFLD definition and risk [15†L425-L430] [15†L455-L458] .
- Prevalence statistics (ADA “Statistics About Diabetes”) [29†L438-L447] , WHO obesity facts [10†L314-L322] , and others cited above.
- Synthetic data literature noting privacy benefits [31†L117-L125] .

Each feature range, disease cutoff, and risk relationship is grounded in these authoritative sources, ensuring our synthetic dataset is clinically accurate and defensible.

